# Carlos Romero Madrid

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**To: The Selection Committee & Group Lead**
Data Analytics and Visualization Group (DAV)
Barcelona Supercomputing Center (BSC-CNS)

## Subject: Application to join the Data Analytics and Visualization Group

Dear Members of the Data Analytics and Visualization Group,

Most resumes begin with a dense list of buzzwords. Let me begin with a sound: the hum of **MareNostrum 5** computing climate turbulence, or the chime of a motion-sensing sphere when my cat **Pipa** knocks it across the floor.

As a **Computational Physicist** (Double Degree Physics & Mathematics, 1st of Class MSc Plasma Physics) and **HPC Scientific Software Engineer** at BSC, I optimize solvers across Europe's tier-0 clusters (MN5, Leonardo, LUMI). Yet my honest conviction is simple: *if petabytes of scientific breakthroughs stay locked behind cold terminal logs and abstract matrices, we have only completed half the mission.*

Inspired by **Domestic Data Streamers**, I treat data as raw physical matter to be touched, heard, and questioned. In my personal studio, I turn this into playful reality:

- **MNusic** (HPC Telemetry Sonification): Translates MareNostrum 5 and LUMI cluster metrics into real-time harmonic soundscapes (Web Audio API) and generative dynamic particles.
- **Pipa-Music-Ball** (Bio-Kinetic Physical Computing): Embedded interactive sphere (ESP32, 6-DoF IMU) translating feline kinetic play into microtonal scales—proof that data interfaces can be playful and tactile.
- **Expo-Curator** (Visual Corpus Analytics & Staging): Open-source application for automated visual curation, catalog design, and exhibition staging for cultural and science spaces.
- **After Effects Lab** (Procedural Motion Design): Photographic composition paired with procedural motion graphics to turn static scientific figures into broadcast-ready visual essays.

I build at lightning speed: pairing with Large Language Models allows me to fluidly bridge C++/Fortran HPC kernels, microcontroller firmware, browser audio synthesis, and visual motion graphics.

Following your flagship initiatives—such as the urban digital twin modeling of **GDU-HPC**, atmospheric narratives of **AIR**, and climate mobility in **REALLOCATE**—I see an organic match with my digital twin work at BSC (DestinationE / ClimateDT). I would be thrilled to join Fernando Cucchietti and the team to co-design the next generation of daring, unforgettable scientific data experiences.

■ **Live Interactive Portfolio & Sound Installation:** Explore real-time MN5 telemetry sonification, day/night audio crossfader, and video cinema mode at: **carrommad.github.io/cool-cv**

Warm regards,


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